





# **Design and Stress-Based Dimensioning of a Two-Way Manual Meat Mincer**

Mechanical Engineering Design Project Report

Preliminary Stress-Based Dimensioning Using Static Concept Sizing

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E/22/401

E/22/410

E/22/413

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# Contents

|          |                                                 |           |
|----------|-------------------------------------------------|-----------|
| <b>1</b> | <b>Conceptualization Details</b>                | <b>1</b>  |
| <b>2</b> | <b>Labeled Picture of the Final Design</b>      | <b>3</b>  |
| <b>3</b> | <b>Failure Modes</b>                            | <b>4</b>  |
| <b>4</b> | <b>Design Data</b>                              | <b>6</b>  |
| <b>5</b> | <b>Assumptions</b>                              | <b>6</b>  |
| <b>6</b> | <b>Material Properties</b>                      | <b>7</b>  |
| <b>7</b> | <b>Design Factors</b>                           | <b>7</b>  |
| <b>8</b> | <b>Design Calculations</b>                      | <b>8</b>  |
| 8.1      | Gear Geometry . . . . .                         | 8         |
| 8.2      | Input Torque and Planetary Gear Ratio . . . . . | 10        |
| 8.3      | Pressure Load . . . . .                         | 10        |
| 8.4      | Crank Arm . . . . .                             | 11        |
| 8.5      | Crank-Arm Deflection . . . . .                  | 12        |
| 8.6      | Grip Pin . . . . .                              | 13        |
| 8.7      | Input Shaft . . . . .                           | 14        |
| 8.8      | Output and Auger Shaft . . . . .                | 15        |
| 8.9      | Gear-Tooth Bending . . . . .                    | 16        |
| 8.10     | Planet Pins . . . . .                           | 18        |
| 8.11     | Barrel Wall . . . . .                           | 19        |
| 8.12     | Grinder Plate . . . . .                         | 20        |
| 8.13     | Four-Arm Grinder Knife . . . . .                | 21        |
| 8.14     | Locking Ring and Front Thread . . . . .         | 23        |
| 8.15     | Upright Support . . . . .                       | 24        |
| 8.16     | Clamp Screw . . . . .                           | 25        |
| 8.17     | Suction Base . . . . .                          | 26        |
| 8.18     | Hopper Sheet . . . . .                          | 27        |
| <b>9</b> | <b>Calculation Summary</b>                      | <b>29</b> |

|                                |           |
|--------------------------------|-----------|
| <b>10 Engineering Drawings</b> | <b>30</b> |
| <b>11 Conclusion</b>           | <b>30</b> |
| <b>12 References</b>           | <b>32</b> |

## List of Figures

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                |    |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|
| 1 | Morphological concept-mapping chart used to generate and combine candidate sub-solutions for the mincer stand, handle, power-transmission system, gripping surfaces, and aesthetic treatment. . . . .                                                                                                                                                                                                                          | 2  |
| 2 | Labeled assembly view of the two-way manual meat mincer. Main components: (1) hand-wheel crank and ring-gear input, (2) planetary gearbox containing the sun, planet, and ring gears, (3) carrier and output shaft, (4) central auger, (5) counter-rotating barrel or cutting assembly, (6) grinder plate and four-arm knife, (7) M72×2 locking ring, (8) upright support and legs, (9) suction base, and (10) hopper. . . . . | 4  |
| 3 | Free-body diagram of the hand-crank input showing the handle force $F_h$ , crank radius $L_h$ , input torque, planetary gearbox, and output-torque path. . . . .                                                                                                                                                                                                                                                               | 8  |
| 4 | Fully dimensioned orthographic engineering drawing of the two-way manual meat-mincer assembly. The drawing should indicate the 16 mm input shaft, 18 mm output and auger shaft, 24/24/72-tooth module-1 planetary gearbox, 4 mm barrel wall, 5 mm grinder plate, and M72×2 locking ring. . . . .                                                                                                                               | 30 |

## List of Tables

|   |                                                                   |    |
|---|-------------------------------------------------------------------|----|
| 1 | Morphological concept-mapping chart . . . . .                     | 1  |
| 2 | Anticipated failure modes and mitigation strategies . . . . .     | 5  |
| 3 | Design data and operating parameters . . . . .                    | 6  |
| 4 | Mechanical properties used for preliminary calculations . . . . . | 7  |
| 5 | Final selected dimensions and calculated margins . . . . .        | 29 |

# 1 Conceptualization Details

The design brief called for a manually operated, two-way (counter-shearing) meat mincer suitable for domestic or small-scale kitchen use.

To generate a broad design space before converging on a single concept, the group constructed a morphological concept-mapping chart. The chart decomposed the product into four independent design functions:

- the stand and handle mechanism;
- the power-transmission system;
- the grip and contact material; and
- the aesthetic treatment.

Multiple candidate solutions were generated for each function. Combinations of these sub-solutions were then evaluated against the main design requirements, including clamping stability, torque multiplication, ergonomics, food safety, and manufacturability.

Table 1: Morphological concept-mapping chart

| Design Function       | Candidate Solutions                  |
|-----------------------|--------------------------------------|
| Handle mechanism      | Paddle, hand wheel, vertical lever   |
| Stand/base mounting   | Suction base, stand claw             |
| Power transmission    | Planetary gear, spur gear, worm gear |
| Grip/contact material | Wood, rubber, plastic                |
| Aesthetic treatment   | Colour scheme, logo design           |

FIGURE 1: CONVERGENCE OF MINCER DESIGN CONCEPTS

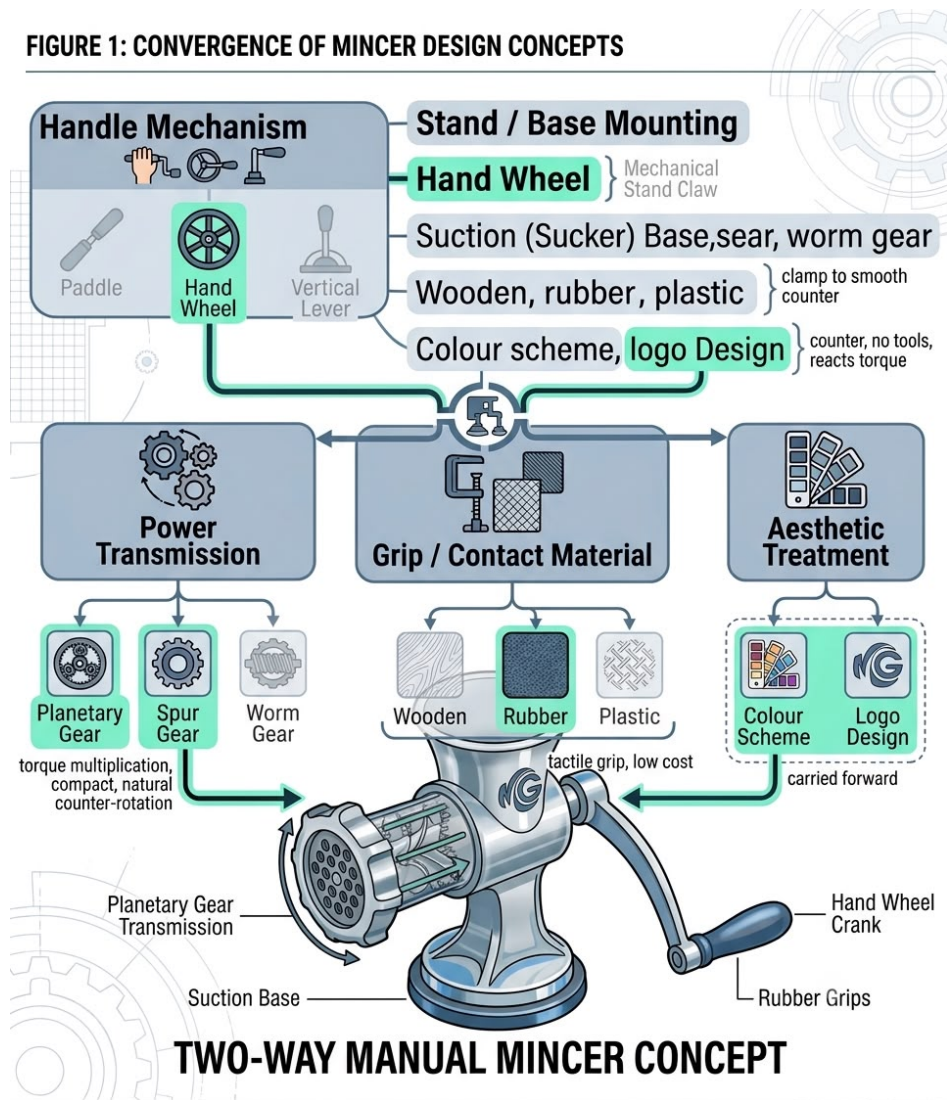


Figure 1: Morphological concept-mapping chart used to generate and combine candidate sub-solutions for the mincer stand, handle, power-transmission system, gripping surfaces, and aesthetic treatment.

Each combination of sub-solutions was screened against the design goals.

A hand-wheel or crank input was favoured over a paddle or vertical lever because it gives the operator a consistent and repeatable moment arm. It is also compatible with a continuously rotating planetary gearbox.

Among the available transmission systems, the **planetary gear train** was selected over the spur-gear and worm-gear alternatives. A fixed-sun, ring-input, carrier-output arrangement provides torque multiplication within a compact and concentric assembly.

The planetary arrangement also supports the desired counter-rotating or two-way shearing action. A single spur-gear or worm-gear pair would require additional gears or transmission stages to obtain the same counter-rotation.

For the stand, a **suction base** was selected over a mechanical stand claw because it can be attached to a smooth countertop without damaging the surface or requiring tools. The

suction base must, however, produce sufficient hold-down force to resist the operating torque. This requirement is checked quantitatively in Section 8.

Rubber was selected for the operator-contact surfaces because it provides good grip and is relatively inexpensive. A coordinated colour scheme and printed logo were selected as the aesthetic treatment.

The resulting concept is therefore a manually cranked, planetary-gear-driven, counter-rotating mincer mounted on a vacuum suction base.

## **2 Labeled Picture of the Final Design**

The converged design integrates a fixed-sun planetary gearbox directly behind the mincing barrel.

The hand crank drives the ring gear, while the planetary carrier provides a torque-multiplied output. This arrangement enables the central auger and outer barrel or cutting assembly to rotate in opposite directions, producing a two-way shearing action.

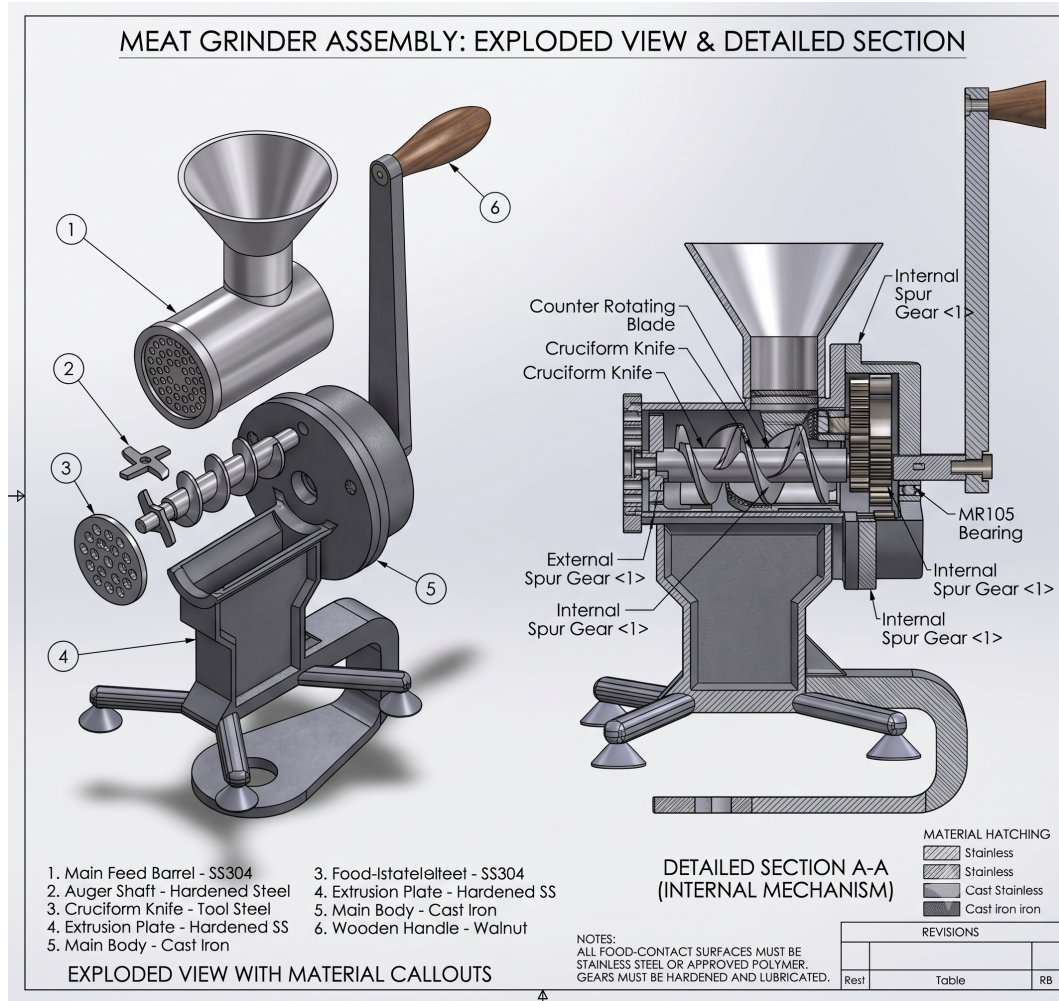


Figure 2: Labeled assembly view of the two-way manual meat mincer. Main components: (1) hand-wheel crank and ring-gear input, (2) planetary gearbox containing the sun, planet, and ring gears, (3) carrier and output shaft, (4) central auger, (5) counter-rotating barrel or cutting assembly, (6) grinder plate and four-arm knife, (7) M72×2 locking ring, (8) upright support and legs, (9) suction base, and (10) hopper.

### 3 Failure Modes

The static stress analysis presented in Section 8 excludes several effects, including gear-tooth fatigue, wear, lubrication breakdown, bone-fragment impact, casting defects, weld defects, and operator misuse.

These effects are considered separately as anticipated failure modes in Table 2.

Table 2: Anticipated failure modes and mitigation strategies

| Component               | Failure Mode                           | Cause                                                                                               | Effect                                                                   | Mitigation                                                                                                     |
|-------------------------|----------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Planetary gear teeth    | Fatigue pitting and wear               | Repeated meshing cycles; the static Lewis check does not account for fatigue or surface durability. | Progressive tooth wear, increased backlash, and eventual tooth fracture. | Surface-harden the module-1 teeth or increase the module and face width for extended service life.             |
| Crank arm and grip pin  | Bending overload                       | Operator applies a force greater than the 100 N design load or misuses the handle.                  | Permanent deformation or fracture near the crank-root section.           | Use the selected 25 mm $\times$ 10 mm root section and avoid excessive handle length.                          |
| Grinder plate and knife | Impact damage                          | Hard bone fragments become trapped between the knife and grinder plate.                             | Local deformation, edge chipping, knife blunting, or jamming.            | Select plate and knife thicknesses based on cutting stiffness and wear resistance, not pressure loading alone. |
| Locking-ring thread     | Thread stripping or loosening          | Repeated tightening, vibration, wear, or incorrect engagement.                                      | Reduction of clamp preload and possible leakage or misalignment.         | Use an M72 $\times$ 2 thread with 12 mm engagement and provide a secure tightening surface.                    |
| Suction base            | Loss of vacuum                         | Worn diaphragm, contamination, air leakage, or an uneven countertop.                                | Base movement or detachment during operation.                            | Use a 120 mm food-safe suction cup and maintain a clean, smooth sealing surface.                               |
| Cast or welded joints   | Porosity, cracks, or incomplete fusion | Manufacturing defects and poor quality control.                                                     | Local stress concentrations and premature structural failure.            | Inspect critical joints and apply appropriate non-destructive testing when required.                           |

## 4 Design Data

Table 3: Design data and operating parameters

| Parameter Description                 | Symbol     | Value  |
|---------------------------------------|------------|--------|
| Internal gauge pressure               | $p$        | 10 kPa |
| Design handle force                   | $F_h$      | 100 N  |
| Effective crank radius                | $L_h$      | 180 mm |
| Grip-pin overhang                     | $L_p$      | 35 mm  |
| Barrel bore diameter                  | $D_b$      | 64 mm  |
| Gear module                           | $m$        | 1 mm   |
| Number of sun-gear teeth              | $N_s$      | 24     |
| Number of planet-gear teeth           | $N_p$      | 24     |
| Number of ring-gear teeth             | $N_r$      | 72     |
| Planet-centre radius                  | $r$        | 24 mm  |
| Handle-force height above base        | $h$        | 170 mm |
| Clamp-force lever arm                 | $L_c$      | 40 mm  |
| Suction-base support radius           | $R_s$      | 110 mm |
| Maintained vacuum pressure difference | $\Delta p$ | 50 kPa |
| Maximum hopper batch mass             | $m_b$      | 5 kg   |
| Corresponding batch weight            | $W_b$      | 49 N   |
| Service/dynamic factor                | $K_s$      | 1.5    |
| Static factor of safety               | $n$        | 3      |

## 5 Assumptions

The following assumptions were used during the preliminary design calculations:

### 1. Static loading

All loads are treated as static or slowly applied. Dynamic amplification caused by sudden jamming, impact, or rapid operator movement is represented approximately by the service factor  $K_s = 1.5$ .

### 2. Ideal planetary load sharing

The transmitted load is assumed to be shared equally among the three planet gears.

### 3. No transmission-loss credit

Gear-mesh and bearing efficiencies are assumed to be 100%. Therefore, the calculated output torque is a conservative upper-bound value for static sizing.

#### 4. Uniform internal pressure

The internal gauge pressure of 10 kPa is assumed to act uniformly on the barrel wall and grinder plate.

#### 5. Excluded degradation mechanisms

Fatigue, wear, corrosion damage, lubrication failure, casting defects, weld defects, and accidental misuse are not included directly in the static stress calculations.

#### 6. Ideal material behaviour

Materials are assumed to be homogeneous, isotropic, and linearly elastic below the yield strength.

#### 7. Simplified stress concentrations

Stress concentrations are ignored unless a specific local factor is introduced, such as the factor used for the grinder-knife arms.

## 6 Material Properties

Stainless steel was selected for the food-contact and major structural components because of its corrosion resistance, cleanability, and suitability for food-processing environments.

For the preliminary static calculations, the nominal material properties listed in Table 4 were adopted.

Table 4: Mechanical properties used for preliminary calculations

| Property                      | Value   |
|-------------------------------|---------|
| Nominal yield strength, $S_y$ | 250 MPa |
| Young's modulus, $E$          | 200 GPa |
| Poisson's ratio, $\nu$        | 0.30    |

## 7 Design Factors

Two independent design factors are used.

- A service or dynamic factor of  $K_s = 1.5$  is applied to the handle-generated torque and bending loads.
- A static factor of safety of  $n = 3$  is applied to the material yield strength.

The allowable normal stress is

$$\sigma_{\text{allow}} = \frac{S_y}{n} = \frac{250}{3} = 83.33 \text{ MPa.} \quad (1)$$

Using the von Mises relationship, the corresponding allowable shear stress is

$$\tau_{\text{allow}} = \frac{\sigma_{\text{allow}}}{\sqrt{3}} = \frac{83.33}{\sqrt{3}} = 48.11 \text{ MPa.} \quad (2)$$

The service factor increases the applied design loads, while the factor of safety reduces the allowable material stress. This combination provides additional protection against uncertain manual loading, jamming, manufacturing tolerances, and simplified analytical models.

## 8 Design Calculations

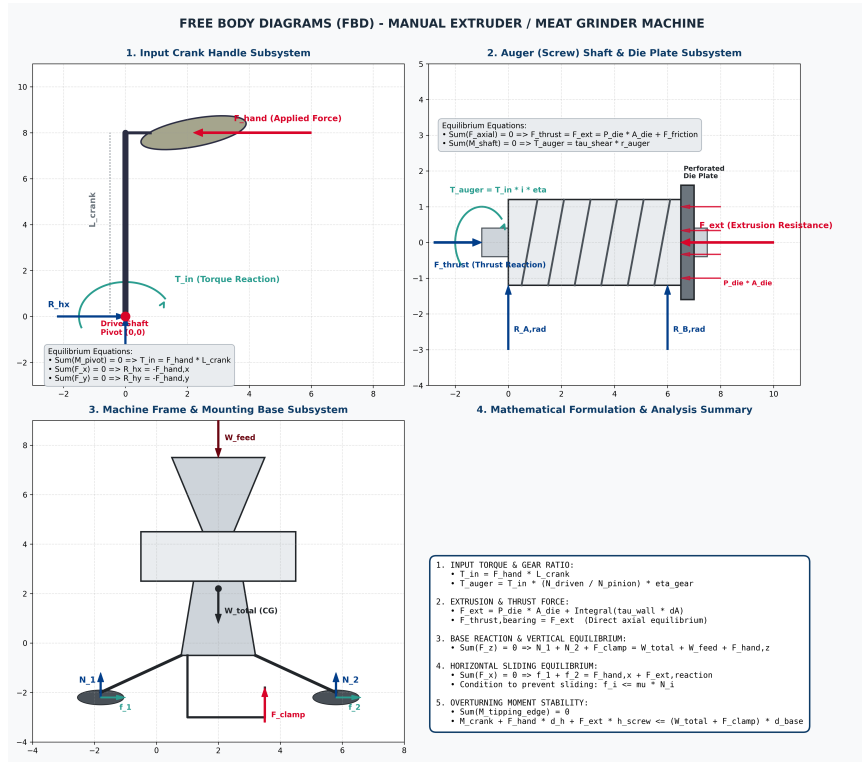


Figure 3: Free-body diagram of the hand-crank input showing the handle force  $F_h$ , crank radius  $L_h$ , input torque, planetary gearbox, and output-torque path.

### 8.1 Gear Geometry

For a gear module of

$$m = 1 \text{ mm,}$$

the selected planetary gear set is

$$N_s = 24, \quad N_p = 24, \quad N_r = 72.$$

The ring-gear condition is satisfied because

$$N_r = N_s + 2N_p = 24 + 2(24) = 72. \quad (3)$$

The pitch diameters are

$$d_s = mN_s = (1)(24) = 24 \text{ mm}, \quad (4)$$

$$d_p = mN_p = (1)(24) = 24 \text{ mm}, \quad (5)$$

$$d_r = mN_r = (1)(72) = 72 \text{ mm}. \quad (6)$$

For standard  $20^\circ$  full-depth involute external teeth, the addendum diameter is

$$d_a = m(N + 2). \quad (7)$$

Therefore, for a 24-tooth sun or planet gear,

$$d_a = 1(24 + 2) = 26 \text{ mm}. \quad (8)$$

The dedendum diameter is approximately

$$d_f = m(N - 2.5). \quad (9)$$

Thus,

$$d_f = 1(24 - 2.5) = 21.5 \text{ mm}. \quad (10)$$

For the internal ring gear, the approximate tooth-tip diameter is

70 mm,

and the approximate root diameter is

74.5 mm.

A practical ring outside diameter of

84 mm

was selected.

## 8.2 Input Torque and Planetary Gear Ratio

The nominal input torque is

$$T_{\text{in}} = F_h L_h. \quad (11)$$

Substituting the design values,

$$T_{\text{in}} = (100)(180) \quad (12)$$

$$= 18\,000 \text{ N mm} \quad (13)$$

$$= 18 \text{ N m}. \quad (14)$$

For a ring-input, fixed-sun, carrier-output planetary arrangement,

$$\frac{\omega_c}{\omega_r} = \frac{N_r}{N_s + N_r}. \quad (15)$$

Therefore,

$$\frac{\omega_c}{\omega_r} = \frac{72}{24 + 72} = 0.75. \quad (16)$$

Assuming ideal power transmission, the output torque is

$$T_{\text{out}} = \frac{T_{\text{in}}}{0.75} = \frac{18}{0.75} = 24 \text{ N m}. \quad (17)$$

Applying the service factor  $K_s = 1.5$ ,

$$T_{\text{in,d}} = 1.5(18) = 27 \text{ N m}, \quad (18)$$

$$T_{\text{out,d}} = 1.5(24) = 36 \text{ N m}. \quad (19)$$

## 8.3 Pressure Load

The barrel bore diameter is

$$D_b = 64 \text{ mm}.$$

The internal gauge pressure is

$$p = 10 \text{ kPa} = 0.01 \text{ N/mm}^2.$$

The axial pressure force acting on the grinder plate is

$$F_p = p \frac{\pi D_b^2}{4}. \quad (20)$$

Therefore,

$$F_p = 0.01 \left[ \frac{\pi (64)^2}{4} \right] \quad (21)$$

$$= 32.17 \text{ N}. \quad (22)$$

This force is relatively small compared with the mechanical loads generated by the crank and gearbox.

## 8.4 Crank Arm

The maximum design bending moment at the crank root is

$$M_d = K_s F_h L_h. \quad (23)$$

Therefore,

$$M_d = 1.5(100)(180) \quad (24)$$

$$= 27\,000 \text{ N mm}. \quad (25)$$

For a rectangular crank arm of width  $b$  and thickness  $t$ , the maximum bending stress is

$$\sigma = \frac{6M_d}{bt^2}. \quad (26)$$

Rearranging for the required thickness,

$$t_{\min} = \sqrt{\frac{6M_d}{b\sigma_{\text{allow}}}}. \quad (27)$$

Using

$$b = 25 \text{ mm},$$

$$t_{\min} = \sqrt{\frac{6(27000)}{25(83.33)}} \quad (28)$$

$$= 8.82 \text{ mm}. \quad (29)$$

A practical crank-arm root section of

$$\boxed{25 \text{ mm} \times 10 \text{ mm}}$$

was selected.

The resulting bending stress is

$$\sigma = \frac{6(27000)}{25(10)^2} \quad (30)$$

$$= 64.8 \text{ MPa}. \quad (31)$$

The resulting yield margin is

$$\text{Yield margin} = \frac{S_y}{\sigma} = \frac{250}{64.8} = 3.86. \quad (32)$$

## 8.5 Crank-Arm Deflection

For a cantilever crank arm,

$$\delta = \frac{F_h L_h^3}{3EI}. \quad (33)$$

The second moment of area is

$$I = \frac{bt^3}{12}. \quad (34)$$

Therefore,

$$I = \frac{25(10)^3}{12} \quad (35)$$

$$= 2083.33 \text{ mm}^4. \quad (36)$$

Using

$$E = 200\,000\text{ N/mm}^2,$$

the crank-tip deflection is

$$\delta = \frac{100(180)^3}{3(200000)(2083.33)} \quad (37)$$

$$\approx 0.47\text{ mm}. \quad (38)$$

## 8.6 Grip Pin

For a grip-pin overhang of

$$L_p = 35\text{ mm},$$

the design bending moment is

$$M_{\text{pin,d}} = K_s F_h L_p. \quad (39)$$

Therefore,

$$M_{\text{pin,d}} = 1.5(100)(35) \quad (40)$$

$$= 5250\text{ N mm}. \quad (41)$$

For a solid circular pin under bending,

$$\sigma = \frac{32M}{\pi d^3}. \quad (42)$$

Hence,

$$d_{\text{min}} = \left( \frac{32M_{\text{pin,d}}}{\pi \sigma_{\text{allow}}} \right)^{1/3}. \quad (43)$$

Substituting,

$$d_{\text{min}} = \left( \frac{32(5250)}{\pi(83.33)} \right)^{1/3} \quad (44)$$

$$= 8.63\text{ mm}. \quad (45)$$

A grip-pin diameter of

$$\boxed{10 \text{ mm}}$$

was selected.

## 8.7 Input Shaft

For a solid circular shaft under torsion,

$$\tau_{\max} = \frac{16T}{\pi d^3}. \quad (46)$$

According to the von Mises criterion,

$$\sigma_{\text{vm}} = \sqrt{3}\tau_{\max}. \quad (47)$$

Therefore, the minimum diameter is

$$d_{\min} = \left( \frac{16\sqrt{3}T}{\pi\sigma_{\text{allow}}} \right)^{1/3}. \quad (48)$$

For the input shaft,

$$T = T_{\text{in,d}} = 27\,000 \text{ N mm}.$$

Thus,

$$d_{\text{in,min}} = \left( \frac{16\sqrt{3}(27000)}{\pi(83.33)} \right)^{1/3} \quad (49)$$

$$= 14.19 \text{ mm}. \quad (50)$$

A practical input shaft diameter of

$$\boxed{16 \text{ mm}}$$

was selected.

The corresponding von Mises stress is approximately

$$\sigma_{\text{vm}} = 58.15 \text{ MPa.}$$

Therefore,

$$\text{Yield margin} = \frac{250}{58.15} = 4.30. \quad (51)$$

## 8.8 Output and Auger Shaft

For the output shaft,

$$T = T_{\text{out,d}} = 36\,000 \text{ N mm.}$$

Therefore,

$$d_{\text{out,min}} = \left( \frac{16\sqrt{3}(36000)}{\pi(83.33)} \right)^{1/3} \quad (52)$$

$$= 15.62 \text{ mm.} \quad (53)$$

A practical output and auger shaft diameter of

$$\boxed{18 \text{ mm}}$$

was selected.

The corresponding torsional von Mises stress is approximately

$$\sigma_{\text{vm}} = 54.45 \text{ MPa.}$$

Thus,

$$\text{Yield margin} = \frac{250}{54.45} = 4.59. \quad (54)$$

The axial stress caused by the pressure load is

$$\sigma_{\text{axial}} = \frac{F_p}{A}. \quad (55)$$

For an 18 mm shaft,

$$A = \frac{\pi(18)^2}{4} \quad (56)$$

$$= 254.47 \text{ mm}^2. \quad (57)$$

Therefore,

$$\sigma_{\text{axial}} = \frac{32.17}{254.47} \quad (58)$$

$$= 0.126 \text{ MPa}. \quad (59)$$

This axial stress is negligible compared with the torsional stress.

The secondary drum shaft carries the design input torque of

$$27 \text{ N m}.$$

Therefore, a

$$\boxed{16 \text{ mm}}$$

secondary drum shaft was selected.

## 8.9 Gear-Tooth Bending

The total tangential force at the ring-gear pitch circle is

$$F_{t,\text{total}} = \frac{2T_{\text{in,d}}}{d_r}. \quad (60)$$

Therefore,

$$F_{t,\text{total}} = \frac{2(27000)}{72} \quad (61)$$

$$= 750 \text{ N}. \quad (62)$$

Assuming equal load sharing among three planet gears,

$$F_{t,\text{mesh}} = \frac{750}{3} = 250 \text{ N}. \quad (63)$$

For a 24-tooth, 20° full-depth involute tooth, the Lewis form factor is approximated by

$$y = 0.154 - \frac{0.912}{N}. \quad (64)$$

Therefore,

$$y = 0.154 - \frac{0.912}{24} \quad (65)$$

$$= 0.116. \quad (66)$$

Using the Lewis bending equation,

$$\sigma_{\text{tooth}} = \frac{F_{t,\text{mesh}}}{b\pi my}. \quad (67)$$

The required face width is

$$b_{\min} = \frac{F_{t,\text{mesh}}}{\sigma_{\text{allow}}\pi my}. \quad (68)$$

Thus,

$$b_{\min} = \frac{250}{(83.33)\pi(1)(0.116)} \quad (69)$$

$$= 8.23 \text{ mm}. \quad (70)$$

A face width of

$$\boxed{12 \text{ mm}}$$

was selected.

The resulting tooth-bending stress is

$$\sigma_{\text{tooth}} = \frac{250}{\pi(1)(12)(0.116)} \quad (71)$$

$$= 57.17 \text{ MPa}. \quad (72)$$

The corresponding yield margin is

$$\text{Yield margin} = \frac{250}{57.17} = 4.37. \quad (73)$$

This Lewis calculation is only a preliminary static bending check. A complete gear design should also consider fatigue, pitting resistance, wear, material hardness, lubrication, manufacturing accuracy, and AGMA design factors.

## 8.10 Planet Pins

The tangential force acting on each planet pin is

$$F_{\text{pin}} = \frac{T_{\text{out,d}}}{3r}. \quad (74)$$

Therefore,

$$F_{\text{pin}} = \frac{36000}{3(24)} \quad (75)$$

$$= 500 \text{ N}. \quad (76)$$

For direct shear,

$$\tau = \frac{4F}{\pi d^2}. \quad (77)$$

Hence,

$$d_{\text{pin,shear}} = \sqrt{\frac{4F_{\text{pin}}}{\pi \tau_{\text{allow}}}}. \quad (78)$$

Therefore,

$$d_{\text{pin,shear}} = \sqrt{\frac{4(500)}{\pi(48.11)}} \quad (79)$$

$$= 3.64 \text{ mm}. \quad (80)$$

For a simply supported pin with a loaded span of

$$L = 12 \text{ mm},$$

the maximum bending moment is approximated by

$$M_{\text{max}} = \frac{F_{\text{pin}}L}{8}. \quad (81)$$

Therefore,

$$M_{\max} = \frac{500(12)}{8} \quad (82)$$

$$= 750 \text{ N mm.} \quad (83)$$

The bending-based minimum diameter is

$$d_{\text{pin,bend}} = \left( \frac{32M_{\max}}{\pi\sigma_{\text{allow}}} \right)^{1/3}. \quad (84)$$

Thus,

$$d_{\text{pin,bend}} = \left( \frac{32(750)}{\pi(83.33)} \right)^{1/3} \quad (85)$$

$$= 4.51 \text{ mm.} \quad (86)$$

A practical planet-pin diameter of

$$\boxed{8 \text{ mm}}$$

was selected.

A carrier-plate thickness of

$$\boxed{6 \text{ mm}}$$

was selected to provide adequate stiffness and bearing support.

## 8.11 Barrel Wall

Treating the barrel as a thin-walled cylinder, the circumferential stress is

$$\sigma_h = \frac{pD_b}{2t}. \quad (87)$$

The theoretical minimum thickness is

$$t_{\min} = \frac{pD_b}{2\sigma_{\text{allow}}}. \quad (88)$$

Therefore,

$$t_{\min} = \frac{0.01(64)}{2(83.33)} \quad (89)$$

$$= 0.00384 \text{ mm}. \quad (90)$$

This theoretical pressure-based thickness is not practical because it does not consider manufacturing, handling, wear, ovalisation, threading, impact, or local attachment loads.

Therefore, a barrel-wall thickness of

$$\boxed{4 \text{ mm}}$$

was selected.

The resulting hoop stress is

$$\sigma_h = \frac{0.01(64)}{2(4)} \quad (91)$$

$$= 0.080 \text{ MPa}. \quad (92)$$

## 8.12 Grinder Plate

For a clamped solid circular plate of radius

$$a = 32 \text{ mm},$$

under uniform pressure, the approximate minimum thickness is

$$t_{\min} = \sqrt{\frac{(3 + \nu)pa^2}{8\sigma_{\text{allow}}}}. \quad (93)$$

Substituting the design values,

$$t_{\min} = \sqrt{\frac{(3 + 0.30)(0.01)(32)^2}{8(83.33)}} \quad (94)$$

$$= 0.225 \text{ mm}. \quad (95)$$

A practical thickness of

$$\boxed{5 \text{ mm}}$$

was selected for cutting stiffness, perforation strength, flatness, and wear resistance.

The approximate unperforated plate stress is

$$\sigma_{\max} = \frac{(3 + \nu)pa^2}{8t^2}. \quad (96)$$

Therefore,

$$\sigma_{\max} = \frac{(3 + 0.30)(0.01)(32)^2}{8(5)^2} \quad (97)$$

$$= 0.169 \text{ MPa}. \quad (98)$$

Applying a conservative stress-concentration factor of 3 for the perforations gives

$$\sigma_{\text{effective}} = 3(0.169) = 0.507 \text{ MPa}. \quad (99)$$

Therefore, cutting stiffness and wear govern the grinder-plate thickness rather than the internal pressure.

### 8.13 Four-Arm Grinder Knife

The torque transmitted by the knife is

$$T_{\text{out,d}} = 36\,000 \text{ N mm}.$$

For four equal knife arms acting at an effective cutting radius of

$$r_k = 28 \text{ mm},$$

the force per arm is

$$F_{\text{arm}} = \frac{T_{\text{out,d}}}{4r_k}. \quad (100)$$

Therefore,

$$F_{\text{arm}} = \frac{36000}{4(28)} \quad (101)$$

$$= 321.43 \text{ N}. \quad (102)$$

Each arm is modelled as a cantilever with

$$l = 22 \text{ mm}, \quad w = 10 \text{ mm},$$

and a local stress factor of

$$K_t = 1.5.$$

The minimum thickness is

$$t_{\min} = \sqrt{\frac{K_t 6F_{\text{arm}} l}{w \sigma_{\text{allow}}}}. \quad (103)$$

Therefore,

$$t_{\min} = \sqrt{\frac{1.5(6)(321.43)(22)}{10(83.33)}} \quad (104)$$

$$= 8.74 \text{ mm}. \quad (105)$$

A knife thickness of

$$\boxed{10 \text{ mm}}$$

was selected.

The resulting bending stress is

$$\sigma_b = \frac{1.5(6)(321.43)(22)}{10(10)^2} \quad (106)$$

$$= 63.64 \text{ MPa}. \quad (107)$$

The corresponding yield margin is

$$\text{Yield margin} = \frac{250}{63.64} = 3.93. \quad (108)$$

A relief angle of approximately

$$\boxed{25^\circ}$$

was selected as a preliminary cutting-geometry value.

## 8.14 Locking Ring and Front Thread

The internal pressure produces only

$$F_p = 32.17 \text{ N}$$

of axial force.

To account for tightening and handling, an assumed axial preload of

$$1 \text{ kN}$$

was used.

Applying the factor of safety,

$$F_{\text{thread,d}} = 3(1000) = 3000 \text{ N.} \quad (109)$$

Using a nominal thread shear diameter of

$$68 \text{ mm,}$$

the required engagement length was estimated as

$$0.584 \text{ mm.}$$

This calculated minimum is extremely small because the applied axial load is low.

A practical thread specification of

$$\boxed{\text{M72} \times 2}$$

with the following dimensions was selected:

- thread engagement length: 12 mm;
- locking-ring outside diameter: 86 mm; and
- locking-ring axial depth: 18 mm.

These dimensions are governed mainly by manufacturability, engagement, alignment, grip, and damage tolerance.

## 8.15 Upright Support

The design bending moment acting on the upright support is

$$M_{\text{support,d}} = K_s F_h h. \quad (110)$$

Therefore,

$$M_{\text{support,d}} = 1.5(100)(170) \quad (111)$$

$$= 25\,500 \text{ N mm}. \quad (112)$$

For a rectangular support of width

$$b = 30 \text{ mm},$$

the required thickness is

$$t_{\min} = \sqrt{\frac{6M_{\text{support,d}}}{b\sigma_{\text{allow}}}}. \quad (113)$$

Therefore,

$$t_{\min} = \sqrt{\frac{6(25500)}{30(83.33)}} \quad (114)$$

$$= 7.82 \text{ mm}. \quad (115)$$

A support section of

$$\boxed{30 \text{ mm} \times 8 \text{ mm}}$$

was selected.

The corresponding bending stress is

$$\sigma = \frac{6(25500)}{30(8)^2} \quad (116)$$

$$= 79.69 \text{ MPa}. \quad (117)$$

The resulting yield margin is

$$\text{Yield margin} = \frac{250}{79.69} = 3.14. \quad (118)$$

## 8.16 Clamp Screw

For a clamp-force lever arm of

$$L_c = 40 \text{ mm},$$

the required clamping force is estimated from moment equilibrium:

$$F_c = \frac{nF_h h}{L_c}. \quad (119)$$

Therefore,

$$F_c = \frac{3(100)(170)}{40} \quad (120)$$

$$= 1275 \text{ N}. \quad (121)$$

An M12 coarse screw has an approximate tensile-stress area of

$$A_t = 84.3 \text{ mm}^2.$$

The direct tensile stress is

$$\sigma = \frac{F_c}{A_t}. \quad (122)$$

Therefore,

$$\sigma = \frac{1275}{84.3} \quad (123)$$

$$= 15.1 \text{ MPa}. \quad (124)$$

An

M12 clamp screw

with a

35 mm

diameter pressure pad was selected.

### 8.17 Suction Base

The overturning moment produced by the handle force is

$$M_o = F_h h. \quad (125)$$

Therefore,

$$M_o = 100(170) \quad (126)$$

$$= 17\,000 \text{ N mm}. \quad (127)$$

At a suction-base support radius of

$$R_s = 110 \text{ mm},$$

the required hold-down force is

$$F_r = \frac{M_o}{R_s}. \quad (128)$$

Therefore,

$$F_r = \frac{17000}{110} \quad (129)$$

$$= 154.55 \text{ N}. \quad (130)$$

Applying the safety factor,

$$F_{r,d} = 3(154.55) \quad (131)$$

$$= 463.64 \text{ N}. \quad (132)$$

The suction force is

$$F = \Delta p A = \Delta p \left( \frac{\pi D^2}{4} \right). \quad (133)$$

Therefore, the required cup diameter is

$$D_{\text{cup,min}} = \sqrt{\frac{4F_{r,d}}{\pi \Delta p}}. \quad (134)$$

Using

$$\Delta p = 50 \text{ kPa} = 0.05 \text{ N/mm}^2,$$

$$D_{\text{cup,min}} = \sqrt{\frac{4(463.64)}{\pi(0.05)}} \quad (135)$$

$$= 108.66 \text{ mm}. \quad (136)$$

A suction-cup diameter of

$$\boxed{120 \text{ mm}}$$

was selected.

A 3 mm metallic backing plate may be used, while the sealing diaphragm should be manufactured from food-safe nitrile rubber or silicone.

## 8.18 Hopper Sheet

The maximum batch mass is

$$m_b = 5 \text{ kg}.$$

Its approximate weight is

$$W_b = 49 \text{ N}.$$

A representative simply supported strip is assumed to have

$$L = 100 \text{ mm}, \quad b = 100 \text{ mm}.$$

Applying the service factor, the design bending moment is estimated as

$$M_d = \frac{K_s W_b L}{4}. \quad (137)$$

Therefore,

$$M_d = \frac{1.5(49)(100)}{4} \quad (138)$$

$$= 1837.5 \text{ N mm.} \quad (139)$$

The required sheet thickness is

$$t_{\min} = \sqrt{\frac{6M_d}{b\sigma_{\text{allow}}}}. \quad (140)$$

Therefore,

$$t_{\min} = \sqrt{\frac{6(1837.5)}{100(83.33)}} \quad (141)$$

$$= 1.15 \text{ mm.} \quad (142)$$

A hopper-sheet thickness of

$$\boxed{1.5 \text{ mm}}$$

with folded edge flanges of approximately

$$\boxed{10 \text{ mm}}$$

was selected.

## 9 Calculation Summary

Table 5: Final selected dimensions and calculated margins

| Design Check                | Calculated Minimum | Selected Value | Calculated Stress         | Yield Margin            |
|-----------------------------|--------------------|----------------|---------------------------|-------------------------|
| Crank-arm thickness         | 8.82 mm            | 10 mm          | 64.8 MPa                  | 3.86                    |
| Input-shaft diameter        | 14.19 mm           | 16 mm          | 58.15 MPa VM              | 4.30                    |
| Output/auger-shaft diameter | 15.62 mm           | 18 mm          | 54.45 MPa VM              | 4.59                    |
| Gear face width             | 8.23 mm            | 12 mm          | 57.17 MPa                 | 4.37                    |
| Planet-pin diameter         | 4.51 mm            | 8 mm           | Below allowable           | > 3                     |
| Barrel-wall thickness       | 0.003 84 mm        | 4 mm           | 0.080 MPa                 | Manufacturing governs   |
| Grinder-plate thickness     | 0.225 mm           | 5 mm           | 0.169 MPa                 | Cutting and wear govern |
| Grinder-knife thickness     | 8.74 mm            | 10 mm          | 63.64 MPa                 | 3.93                    |
| Upright-support thickness   | 7.82 mm            | 8 mm           | 79.69 MPa                 | 3.14                    |
| Suction-cup diameter        | 108.66 mm          | 120 mm         | Hold-down force at 50 kPa | Approximately 3         |
| Hopper-sheet thickness      | 1.15 mm            | 1.5 mm         | Below allowable           | > 3                     |

All stress-governed selected dimensions provide a yield margin equal to or greater than the target factor of safety of 3.

The barrel wall and grinder plate have extremely small theoretical pressure-based minimum thicknesses. Their selected dimensions are therefore governed by manufacturability, stiffness, wear resistance, handling, perforations, threads, and local loading rather than internal pressure alone.

## 10 Engineering Drawings

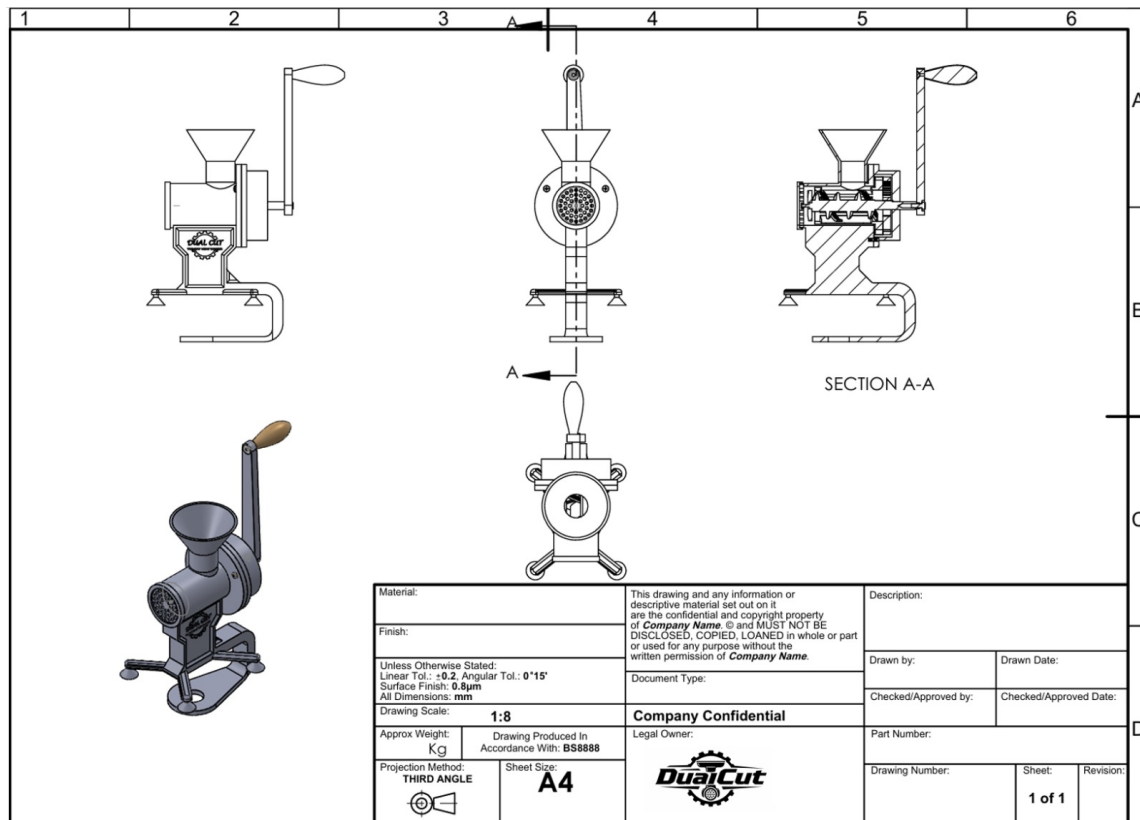


Figure 4: Fully dimensioned orthographic engineering drawing of the two-way manual meat-mincer assembly. The drawing should indicate the 16 mm input shaft, 18 mm output and auger shaft, 24/24/72-tooth module-1 planetary gearbox, 4 mm barrel wall, 5 mm grinder plate, and M72 $\times$ 2 locking ring.

## 11 Conclusion

A preliminary stress-based design was completed for a manually operated, two-way meat mincer using a planetary gear transmission.

The selected planetary arrangement provides a theoretical output torque of 24 N m from a nominal 18 N m crank input. After applying a service factor of 1.5, the design input and output torques are 27 N m and 36 N m, respectively.

The selected shaft, crank, gear, knife, support, and suction-base dimensions satisfy the preliminary static factor-of-safety requirement of 3. The smallest calculated yield margin is approximately 3.14 for the upright support.

Several dimensions, particularly the barrel wall and grinder plate, are controlled by manufacturing, cutting stiffness, wear, food-safety, and handling requirements rather than pressure stress.

Further design development should include:

- detailed CAD modelling and assembly-interference checking;

- AGMA-based gear fatigue and contact-stress calculations;
- bearing selection and bearing-life calculations;
- key, spline, or coupling design;
- finite-element analysis of the crank, carrier, knife, and support structure;
- food-contact material and surface-finish verification;
- prototype manufacture and experimental torque testing; and
- cleaning, disassembly, and hygiene assessment.

## 12 References

### References

- [1] Budynas, R. G. and Nisbett, J. K. (2020). *Shigley's Mechanical Engineering Design*. 11th ed. McGraw-Hill Education.
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